

Undekoti Rowan Ronald

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PROFESSIONAL SUMMARY

Machine Learning Engineer graduating with an M.Tech (Integrated) in Artificial Intelligence from SRM Institute of Science and Technology (CGPA 8.47/10.0). Experienced in deep learning, computer vision, NLP, and self-supervised and generative AI models, with hands-on work in PyTorch, TensorFlow, and Hugging Face Transformers. Skilled at taking models from research to real-time, edge-optimized deployment on devices such as the Raspberry Pi 5. Enjoys building AI-powered applications and reliable backend systems, with working knowledge of Java, SQL, and Linux.

EDUCATION

SRM Institute of Science and Technology

M.Tech (Integrated) in Artificial Intelligence; CGPA: 8.47/10.0

Chennai, India

Jul 2026

EXPERIENCE

Valeo India Private Limited

R&D AI Intern

Chennai, India

Oct 2025 – Present

- Architected a real-time smart helmet dashboard system deployed on **Raspberry Pi 5**, engineering an OpenVINO-accelerated, multiprocessing pipeline (PySide6, GStreamer) with **YOLOX Nano** and **YuNet/SFace** for on-device object and face detection.
- Built and maintained a second, simplified deployment branch using **ONNX Runtime**, **InsightFace**, and OpenCV for cross-platform debugging, keeping the detection pipeline portable outside the primary Raspberry Pi 5 target.
- Built AI workflow automation tools using LLMs, **Retrieval-Augmented Generation (RAG)**, LangChain, and **Model Context Protocol (MCP)** integrations, reducing manual tasks by 30%.
- Led software validation initiatives, reducing deployment-critical bugs by 20% through systematic testing protocols.

PROJECTS

Deepfake Audio Detection Framework | PyTorch, HuBERT, XLS-R, Knowledge Distillation

Jan 2026 – Present

- Built a modular deepfake audio detection pipeline combining HuBERT and XLS-R, reaching **0.72% Equal Error Rate (EER)** on ASVspoof 2019, outperforming multilingual baselines by 15%.
- Applied knowledge distillation to produce DistilHuBERT (40% parameter reduction) with 13ms latency for mobile/edge deployment.
- Designed an automated feature-extraction pipeline processing 250+ hours of audio data.

Efficient LLM Fine-Tuning Framework | PyTorch, Hugging Face Transformers, QLoRA

Feb 2025

- Built a fine-tuning pipeline using QLoRA with **4-bit NormalFloat (NF4) quantization**, cutting trainable parameters by 98% and memory usage by 60% on a single GPU.
- Tuned gradient accumulation to hold 94% accuracy on GSM8K while reducing training time by 40%.

Guard My Click: Phishing Detection Web App | Python, Flask, XGBoost, Google API

Dec 2022 – Feb 2023

- Built a hybrid **XGBoost** + fuzzy-logic phishing classifier and Flask web app, reaching 95% accuracy on a 50,000-email dataset via the Gmail API.

RESEARCH & PUBLICATIONS

Book Chapter: Undekoti, R. R. et al. (2026). “Mathematical Reasoning Enhancement Through Fine-tuning: A Comparative Analysis of Gemma-2-2B-IT on GSM8K.” In *Advanced Pathways in Electrical, Communication, and Automation*. CRC Press. ISBN: 9781032867915.

Research Paper: “Monolingual vs. Multilingual Self-Supervised Models for Deepfake Audio Detection: A Comparative Analysis of HuBERT and XLS-R on ASVspoof 2019.” *In preparation, March 2026*.

TECHNICAL SKILLS

Languages: Python, Java, C, SQL

Core CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming

Machine Learning & AI: Machine Learning, Deep Learning, Computer Vision, NLP, LLMs, Retrieval-Augmented Generation (RAG), Generative AI, Prompt Engineering, Knowledge Distillation, Model Quantization

Frameworks & Libraries: PyTorch, TensorFlow, Keras, Hugging Face Transformers, PEFT/LoRA, LangChain, Scikit-Learn, OpenCV, Flask, React

Specializations: YOLOX, YOLOX Nano, MobileNet, FaceNet, QLoRA, 4-bit Quantization, Knowledge Distillation, Edge AI Optimization

Backend & Databases (coursework): Core Java, JDBC, MySQL, REST APIs, Spring Boot fundamentals

Cloud & Tools: AWS (SageMaker, EC2), Docker, Git, GitHub, Linux, Raspberry Pi 5, Pandas, NumPy